HiDMamba: A Hierarchical Distillation Cascade for Interpretable, Edge-Deployable Mamba-Based ECG Classification

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1. Abstract

Automated ECG diagnosis is entering clinical and wearable settings faster than trust can be established, and the models driving this shift are largely black boxes. Cardiologists need diagnostic support that is both accurate across the full range of multi-label conditions and small enough to run at the point of care. Automated ECG interpretation faces a tradeoff. Models accurate enough for multi-label diagnosis are usually too large and too opaque for edge deployment. Models small enough to deploy rarely explain their predictions.

I am presenting **HiDMamba**, a four-tier hierarchical knowledge distillation cascade: **DEAN**, **HOD**, **Professor**, and **Chief Resident**. It mirrors clinical coarse-to-fine reasoning and produces one lightweight, standalone model. **DEAN** is a Mamba-based state space model, pretrained with masked signal reconstruction and lead-wise contrastive learning on PTB-XL (21,837 12-lead ECGs), then fine-tuned on its five-superclass, twenty-three-subclass hierarchy. Knowledge passes through two teacher tiers, five superclass **HODs** and twenty-three subclass **Professors**. A final skip-distillation stage compresses the cascade into a sub-1M-parameter **Chief Resident** that runs inference in under 10ms, using sigmoid-based logit distillation from Professors, a gating-mimicry loss for superclass triage, and relational distillation of DEAN's state-space dynamics. A cross-attention consultation module over Professor embeddings is used only during training and dropped at inference, so the deployed model needs no teacher at runtime. For interpretability, a FAISS-indexed prototype retrieval layer over DEAN's embedding space surfaces similar historical ECGs alongside each prediction.

2. Introduction

2.1 Motivation

Automated 12-lead ECG interpretation pulls a model in three directions at once. Accuracy requires picking up on subtle, multi-lead patterns across many possible diagnoses, most of which are rare and can occur together in the same recording. Trust requires that a prediction come with a reason a physician can actually check, not just a probability. Deployability, especially for wearables and point-of-care devices, requires a model small and fast enough to run on limited hardware. In practice, models built for accuracy tend to be too large to deploy. Models built to be small are usually trained from scratch, so they miss out on the benefits of large-scale pretraining. Almost nothing addresses the trust requirement directly.

PTB-XL [1] captures this tension well. It has 21,837 twelve-lead recordings, enough data to support self-supervised pretraining, and its labels follow a five-superclass, twenty-three-subclass hierarchy, where a single recording can carry several diagnoses at once across both levels. Most existing benchmarks either flatten this hierarchy or only use part of it. The original PTB-XL benchmarking study [2] sets strong baselines across architectures, but treats each task as one flat classification problem and reports AUC alone. Multi-teacher distillation methods built for ECG, like HMT-KD [16], show that distilling from several teachers into one lightweight student helps with deployability. But they group their teachers by model capacity, not by the diagnostic structure the dataset already gives us.

This leaves an open question: if we distill along the actual clinical hierarchy, superclass first and then subclass, the same way a clinician narrows down a diagnosis, does that give us a better-justified and more interpretable path to a deployable model than grouping teachers by capacity does?

2.2 Related Work

ECG deep learning and benchmarking. PTB-XL [1] is the reference dataset for this work. Strodthoff et al. [2] built the first systematic cross-architecture benchmark on it and found that convolutional architectures (ResNet and Inception style) performed best across six diagnostic and metadata prediction tasks. That benchmark reports only macro-averaged AUC and no macro-F1 figure, a distinction this proposal keeps in mind throughout (see Working Method and Evaluation).

State space models for physiological time series. Mamba [3] introduced selective, input-dependent state space models as a near-linear-time alternative to attention for long sequences. Two unrelated papers share the name "MedMamba." Yue & Li [4] apply a vision-Mamba hybrid to 2D medical image classification. He et al. [5] independently propose a Mamba variant built specifically for medical time series such as ECG and EEG, with inductive biases for spatial centralization, multi-timescale composition, and non-causal dependency. The latter is the architecturally relevant precedent for DEAN, and the two papers are cited separately throughout to avoid the confusion present in earlier drafts of this proposal. Qiang et al.'s ECGMamba [6] applies a bidirectional SSM directly to ECG classification, with no distillation component. Ali et al. [7] show that selective SSMs can be reformulated as implicit attention, a result this work reuses both for relational distillation and for the interpretability overlay.

Self-supervised pretraining for ECG. 3KG [8] introduced physiologically motivated augmentations for contrastive ECG pretraining. Mehari & Strodthoff [9] benchmarked several contrastive frameworks (SimCLR, BYOL, CPC-style) directly on 12-lead ECG.

CoRe-ECG [10] is the most direct motivation for this work's hybrid objective: it reports that combining contrastive and reconstructive pretraining outperforms either alone on PTB-XL.

Knowledge distillation, from single-teacher to hierarchical. Distillation through temperature-softened targets originates with Hinton et al. [11]. Relational Knowledge Distillation [12], which matches pairwise (Gram-matrix) structure instead of raw activations, is the specific mechanism this work adapts to work around the basis non-uniqueness of Mamba's internal state. For multi-label settings specifically, Yang et al. [13] and, separately, Zhang, Qin et al. [14] both show that naive softmax-based logit distillation fits multi-label targets poorly, which is why this work uses per-label sigmoid distillation throughout. Mirzadeh et al.'s Teacher Assistant Knowledge Distillation [15] shows that student performance drops when the teacher-student capacity gap is too large, and that chaining intermediate-sized networks helps close that gap. This directly motivates structuring the DEAN-to-Chief-Resident transfer as three hops instead of one. HMT-KD [16] is the closest prior ECG-specific system, distilling from three differently sized teachers into one student. This work differs in two ways: each teacher is scoped to a real ontological subset of the label space, superclass or subclass, rather than differentiated purely by capacity, and the cascade is paired with an interpretability layer, which HMT-KD's published abstract does not describe.

Prototype-based interpretability and retrieval. ProtoECGNet [17] is the direct precedent for prototype-based, case-grounded explanation on PTB-XL's full multi-label task. This work builds a comparable retrieval mechanism over DEAN's embedding space, rather than a jointly trained end-to-end prototype network. Retrieval runs through HNSW [19] as implemented in FAISS, on CPU

Exploiting hierarchical label structure directly. Wang & Hauskrecht [18] model PTB-XL's five-superclass, twenty-three-subclass dependency structure explicitly, using a conditional tree-structured Bayesian network. This is the closest existing precedent for treating the label hierarchy itself as a first-class modeling object, which this work does through distillation rather than a graphical model.

2.3 Preliminary Concepts

State Space Models / Mamba. A selective SSM maps an input sequence to an output sequence through a continuous-time-inspired recurrence with input-dependent parameters. This gives near-linear-time sequence modeling without attention's quadratic cost [3]. Its hidden state has no canonical basis, which is the technical reason this work uses relational rather than pointwise feature distillation.

Knowledge Distillation. A compact student is trained to match a temperature-softened version of a larger teacher's output distribution, in addition to the ground truth [11]. This transfers "dark knowledge" beyond what hard labels alone provide.

Relational Knowledge Distillation (RKD). Instead of matching teacher and student representations directly, which requires matching dimensionality and a shared basis, RKD matches the pairwise relational structure between representations, such as a Gram or distance matrix. This structure stays invariant regardless of the specific basis or dimensionality of either representation [12].

Contrastive Learning / InfoNCE. A representation is trained so that augmented views of the same instance, the positives, are pulled together in embedding space, while other instances in the batch, the negatives, are pushed apart. This is formalized through the InfoNCE objective [23]. The momentum-queue mechanism [22] decouples the effective negative pool size from the physical batch size, which this work relies on given Kaggle's VRAM ceiling.

Homoscedastic Uncertainty Weighting. Multiple loss terms with different scales and units can be combined using learned per-task log-variance parameters instead of fixed manual weights [21]. This work uses that to balance the masked-reconstruction and contrastive losses during DEAN pretraining.

3. Working Methodology

3.1 Workflow Overview

Stage 1 : Data and hierarchy construction. Parse PTB-XL's SCP-statement metadata into an explicit 5-superclass × 23-subclass hierarchy map; resample all recordings to 100Hz; apply the official patient-stratified fold split (1-8 train / 9 val / 10 test); compute per-lead normalization statistics.

Stage 2 : DEAN self-supervised pretraining. Train the Mamba-based DEAN encoder jointly on masked signal reconstruction (MSE) and lead-wise contrastive learning (InfoNCE with a momentum queue), combined via learned uncertainty weights per Kendall et al. [21]:

$$L_{DEAN} = \tfrac{1}{2}\cdot\exp(-s_{MSE})\cdot L_{MSE} + \tfrac{1}{2}\cdot\exp(-s_{InfoNCE})\cdot L_{InfoNCE} + \tfrac{1}{2}(s_{MSE} + s_{InfoNCE})$$

Stage 3 : DEAN supervised fine-tuning. Linear-probe the frozen encoder on the 23-subclass task to validate representation quality, then unfreeze and fine-tune end-to-end; DEAN is frozen from this point forward and used only as a teacher.

Stage 4 : Gating Network training. Train a 264K-parameter MLP on frozen DEAN embeddings to predict superclass routing probabilities g(x), supervised against ground-truth superclass labels plus an entropy regularizer discouraging collapse.

Stage 5 : HOD distillation. For each of the 5 superclasses, train a HOD via per-label sigmoid

distillation from DEAN, Gram-matrix RKD on DEAN's SSM output sequence, and a cross-superclass contrastive term (DEAN-anchored, pushing distinct HODs' representations apart), plus ordinary supervised loss.

Stage 6 : Professor distillation. For each of the 23 subclasses (grouped and trained by parent superclass), train a Professor from its parent HOD using the same distillation structure as Stage 5, plus a hierarchy-consistency term preventing a subclass probability from exceeding its parent superclass probability.

Stage 7 : Teacher-output caching. With DEAN, Gating, all HODs, and all Professors now frozen, precompute and persist to disk: Professor per-subclass logits, Professor penultimate embeddings, Gating routing distributions g(x), and DEAN's SSM output sequences — for every training-set sample, once.

Stage 8 : Chief Resident skip-distillation. Train the standalone Chief Resident (patchifier → single bidirectional Mamba block → classification head) against the cached artifacts from Stage 7, using the corrected three-term loss:

> **$L_{ChiefResident} = L_{supervised} + \lambda 1 \cdot L_{professor_KD} + \lambda 2 \cdot L_{gate_mimic} + \lambda 3 \cdot L_{RKD}$**

During training only, a Cross-Attention Consultation module (query = student's own representation, key/value = cached Professor embeddings) produces an auxiliary, richer prediction; a self-distillation term trains the standalone (Consultation-free) path to match it. At the end of Stage 8, the Consultation module and its auxiliary head are discarded; only the patchifier, Mamba block, and standalone head are retained as the deployed artifact.

Stage 9 : Prototype index construction. Independently of Stages 4-8, extract frozen DEAN embeddings for the full training set and build a CPU-resident FAISS `IndexFlatIP` for nearest-neighbor prototype retrieval.

Stage 10 : Threshold calibration and evaluation. Grid-search per-label decision thresholds on the validation fold to maximize F1; evaluate the full standalone Chief Resident (plus the offline prototype lookup) on the held-out test fold exactly once; benchmark both core-model and end-to-end latency.

3.2 HiDMamba Pipeline

Input: 12-lead ECG recordings from PTB-XL, with SCP-statement diagnostic metadata

Output: Multi-label diagnostic prediction with per-label confidence, accompanied by retrieved prototype ECGs as explanation, from a standalone deployable Chief Resident model

Step 1:

1. Parse PTB-XL's SCP-statement metadata into a 5-superclass × 23-subclass hierarchy map

2. Resample all recordings to 100Hz
3. Apply the official patient-stratified fold split (1-8 train / 9 val / 10 test)
4. Compute per-lead normalization statistics

Step 2:

1. Mask segments of the input ECG signal
2. Compute masked-signal reconstruction loss (MSE) on the Mamba-based DEAN encoder
3. Generate augmented lead-wise views and compute contrastive loss (InfoNCE with a momentum queue)
4. Combine both losses via learned uncertainty weights → L_DEAN

Step 3:

1. Linear-probe the frozen DEAN encoder on the 23-subclass task to check representation quality
2. Unfreeze and fine-tune DEAN end-to-end
3. Freeze DEAN; retain it as a teacher only for all following stages

Step 4:

1. Extract frozen DEAN embeddings for all training samples
2. Train a 264K-parameter MLP (Gating Network) to predict superclass routing probabilities g(x)
3. Supervise against ground-truth superclass labels with an entropy regularizer to discourage collapse

Step 5: (repeat for each of the 5 superclasses)

1. Distill per-label sigmoid targets from DEAN
2. Apply Gram-matrix RKD on DEAN's SSM output sequence
3. Apply a cross-superclass contrastive term to push distinct HODs' representations apart
4. Add the supervised loss on ground-truth labels → train HOD

Step 6: (repeat for each of the 23 subclasses, grouped by parent superclass)

1. Distill per-label sigmoid targets, Gram-matrix RKD, and contrastive term from the parent HOD (same structure as Step 5)
2. Add a hierarchy-consistency term so a subclass probability cannot exceed its parent superclass probability → train Professor

Step 7:

1. Freeze DEAN, Gating Network, all HODs, and all Professors
2. Precompute and store to disk, once, for every training sample: Professor per-subclass logits, Professor penultimate embeddings, Gating routing distributions g(x), DEAN's SSM output sequences

Step 8:

1. Build the Chief Resident: patchifier → single bidirectional Mamba block → classification head
2. Compute supervised loss against ground-truth labels
3. Compute Professor-KD loss (temperature-scaled sigmoid distillation) from cached Professor logits
4. Compute gate-mimicry loss from cached Gating distributions
5. Compute RKD loss from cached DEAN SSM output sequences
6. Run a Cross-Attention Consultation module (query = student representation, key/value = cached Professor embeddings) to produce an auxiliary richer prediction
7. Compute self-distillation loss so the standalone path matches the Consultation-augmented path
8. Sum all terms → L_ChiefResident and backpropagate
9. After training, discard the Consultation module and auxiliary head; keep only the patchifier, Mamba block, and standalone head as the deployed model

Step 9:

1. Extract frozen DEAN embeddings for the full training set
2. Build a CPU-resident FAISS `IndexFlatIP` for nearest-neighbor prototype retrieval

Step 10:

1. Grid-search per-label decision thresholds on the validation fold to maximize F1
2. Evaluate the standalone Chief Resident (with prototype lookup) on the held-out test fold, once
3. Benchmark core-model latency and end-to-end latency

Step 11 (deployment / inference):

1. Capture new ECG sample
2. Pass it through the deployed Chief Resident (patchifier → Mamba block → classification head) to get per-label probabilities
3. Apply calibrated thresholds → diagnostic labels
4. Query the FAISS index with the sample's DEAN embedding → retrieve nearest prototype ECGs
5. Return prediction with prototype-based explanation
6. Repeat for next incoming sample

3.3 Dataset Used

PTB-XL is publicly available 12-lead ECG dataset hosted on PhysioNet.It contains 21,837 12-lead ECG recordings from 18,884 patients, covering a wide range of cardiac conditions with annotations for more than 60 diagnostic statement categories, including arrhythmias, myocardial infarction, and other heart diseases, originally sampled at 1000 Hz. PhysioNet distributes the signals at both the original 500Hz and a downsampled 100Hz rate, which is what this project uses.

3.4 Evaluation Metrics

Primary metrics

1. **Macro-averaged F1 -** the headline metric, computed per label and averaged across all labels. This matters most for PTB-XL because the label distribution is heavily imbalanced across superclasses and especially subclasses, so F1 is a more honest signal than accuracy or AUC alone would be.
2. **Macro-averaged AUC (AUROC)** - computed per label and averaged, kept specifically so results are directly comparable to Strodthoff et al.'s benchmark [2], which reports only macro-AUC and no F1 figure. Reporting both AUC and F1 closes that gap rather than picking a different metric and losing comparability.

Supporting procedure that feeds into F1

3. **Per-label threshold calibration** - not a metric itself, but the step that makes macro-F1 meaningful for a multi-label sigmoid output. Decision thresholds are grid-searched per label on the validation fold to maximize F1, then frozen and applied once to the held-out test fold (Stage 10), so the test number isn't threshold-tuned on test data.

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